



Create Cute Apps With Qt

This article covers application development with Qt SDK and database connectivity for simple applications.

If you plan to develop eye-candy applications or some nice games, or if you need to create applications to store and retrieve data at home or at office, Qt is one of the best solutions for you.

A little background on Qt

Qt, as you perhaps already know, is a cross-platform framework used to create rich GUI applications. It's most popularly known as playing a major role in KDE applications and libraries. One of its primary advantages is that it is available for many platforms, including Linux, Macintosh and Windows. It even has its wings spread in the mobile development space with ports available for embedded Linux (Qt Extended, previously known as Qtopia), Symbian (about to release) and Windows CE.

Qt is not just restricted to GUI applications but comes with many features like multi-threading, networking, database connectivity, 2D/3D graphics with OpenGL, XML, SVG, Phonon support, WebKit integration, etc.

Although Qt programming is C++ based, you don't have to be disappointed by this fact if you are a fan of Java, Python, Ruby

or other languages. Official Java bindings are available as Qt Jambi and third-party bindings are available for many languages like Python, Ruby, etc. It can be easily integrated with popular IDEs like Eclipse and with the recent release of Qt Creator, application development is ultra simple.

Qt was started by Trolltech and was acquired by Nokia last year. Nokia later renamed it Qt Development Frameworks. Qt comes under a multi-licence model that allows both open source (under GPLv3, LGPLv2.1) and commercial usage.

Signals and slots

Qt event handling is mainly based on a signals and slots mechanism that allows communication between different objects (widgets). When an event occurs, a signal is emitted from the source widget, triggering a slot on the target widget as per the registered mapping.

A slot is nothing but a function holding code for event handling. For example, when you click the *Exit* button, a signal is emitted, which is received by the main window and the *close* slot is executed to quit the application. Many signals and slots are predefined for all widgets, and we can always add custom signals and slots to any widget.